

Total No. of Questions : 4]

SEAT No. :

PD-396

[Total No. of Pages : 2

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B.E. (Artificial Intelligence and Data Science) (Insem.)

DEEP LEARNING (Elective - V)

(2019 Pattern) (Semester -VIII) (417532 - D)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Attempt Q. No. 1 or Q. No. 2 and Q. No. 3 or Q. No. 4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

Q1) a) Enlist different optimization algorithm which is commonly used in deep learning? Describe any one. [5]

b) Describe Advantage & challenges of Deep Learning? [5]

c) How Deep Learning works? Explain it? [5]

OR

Q2) a) Enlist different hyper parameter in Deep Learning? Explain any two? [5]

b) Describe Vanishing gradient & exploding gradient? [5]

c) Explain different application of deep learning. [5]

Q3) a) Explain perceptron & multi-layer perceptron? [5]

b) Define loss functions? Explain with reference to classifications. [5]

c) Explain the concept of biological neuron with comparison of artificial neuron. [5]

OR

P.T.O.

- Q4)** a) Illustrate the need of Activation function in Neural Networks? [5]
- b) Define loss functions? Explain with reference to regression. [5]
- c) How does a neural network get trained? [5]



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